Examiners' commentaries 2023 (May)

ST2134 Advanced statistics: statistical inference

Important note

This commentary reflects the examination and assessment arrangements for this course in the academic year 2022–23. The format and structure of the examination may change in future years, and any such changes will be publicised on the virtual learning environment (VLE).

Information about the subject guide and the Essential reading references

Unless otherwise stated, all cross-references will be to the latest version of the subject guide (2021). You should always attempt to use the most recent edition of any Essential reading textbook, even if the commentary and/or online reading list and/or subject guide refer to an earlier edition. If different editions of Essential reading are listed, please check the VLE for reading supplements – if none are available, please use the contents list and index of the new edition to find the relevant section.

General remarks

Learning outcomes

At the end of this half course and having completed the essential reading and activities you should be able to:

- explain the principles of data reduction
- judge the quality of estimators
- choose appropriate methods of statistical inference to tackle real problems.

Key steps to improvement

Candidates should pay close attention to the essential topics of sufficient statistics, point estimation, interval estimation and hypothesis testing. Being able to state and prove key theorems and lemmas simply requires close study of these in the subject guide. The point estimation techniques of method of moments estimation and maximum likelihood estimation are standard (and are even introduced in **ST104B Statistics 2**), with questions typically varying in terms of the probability distribution from which the random sample is drawn (with many common distributions covered in **ST104B Statistics 2** and **ST2133 Advanced statistics: distribution theory**). Pivotal functions feature heavily in Chapter 4 (Interval estimation), and likelihood ratio tests similarly feature heavily in Chapter 5 (Hypothesis testing).

An examination paper attempts to cover as much of the syllabus as possible, but inevitably some parts are left out. As the included parts will vary from year to year it is not enough to practise only on the past papers. Candidates should practise on all the examples, activities and sample examination questions in the subject guide in order to understand the relevant theory and adequately prepare for the examination. Afterwards you may try one or two past papers for further practice.

The examination paper for this course involves four questions and each of the questions contains several parts which contribute towards the final grade with the different numbers of marks reflecting their difficulty. The difficulty of each part is usually assessed after taking into account the level of thinking related to statistical inference and the extent of algebraic computations required. Take 5–10 minutes to read the whole paper and try to identify the difficulties of each question in comparison to your ability. You will find that some questions could be much easier for you than others.

If you are unable to solve an exercise due to algebraic computations or lack of knowledge of the material of **ST2133 Advanced statistics: distribution theory**, do not abandon an exercise. Write clearly the procedure which should be followed for the relevant statistical inference task(s). Remember that the primary aspect you are being examined on is the ability to *understand* the statistical methodology.

A good knowledge of basic calculus, such as the ability to compute integrals, sums and derivatives, will give you invaluable help for completing the examination paper. Also, the main concepts of **ST2133 Advanced statistics: distribution theory** (random variables and their distribution, expectation, independence, conditional expectation etc.) should be well-understood before attempting this half course. If you are not comfortable with any of these, go back and revise. Memorising basic properties such as the moments and probability mass/density functions of some standard distributions (such as the binomial, Poisson, uniform, normal and gamma) is also a good idea.

When you are trying to prove an equality or inequality be careful of what is on the left-hand and right-hand sides. Justify the steps in the calculations as much as possible. If you cannot get a question right, do not just write in the correct answer without any further explanation. If you just write on the script what you think went wrong you will get more marks. Also, try to avoid writing more than you are asked to write. For example, when asked to *state* a theorem, do not give the proof as well as this simply wastes time.

Be clear in your answers. When giving the probability mass/density function of a random variable you should also state its sample space (also known as its support). When you are deriving an estimator or a sufficient statistic also state the parameter to which it corresponds. Sometimes it is a good idea to underline your final result (if applicable).

Make sure that you are able to provide the definitions of basic concepts such as estimators vs. estimates, confidence intervals and hypothesis tests. Remember that hypotheses are statements about population parameters which are estimated using functions of the data (that is, *statistics*). In the relevant calculations you should always be able to distinguish a statistic from a parameter and be clear on what they represent, respectively.

Practise operations involving products and sums. A substantial number of candidates fail to write them down correctly. Quite often they omit or write the indices incorrectly which inevitably leads to errors. A typical example of this is the following:

Let $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ be a random sample from a population with probability density function $f_{X_i}(x_i; \theta)$, often written for simplicity as $f_X(x; \theta)$. An observed sample is $\mathbf{x} = \{x_1, x_2, \ldots, x_n\}$. The fact that this sample is a random sample indicates that we can write the joint probability density function of $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ as:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^{n} f_{X_i}(x_i; \theta).$$

Make sure that you write the above and *not* the following:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^{n} f_X(x; \theta)$$

which can lead to the *incorrect* expression:

$$f_{\mathbf{X}}(\mathbf{x};\theta) = (f_X(x;\theta))^n.$$

Note that if you were asked to provide the log-likelihood function $l_{\mathbf{X}}(\theta;\mathbf{x})$ the calculations would be formed of *sums* rather than *products* (refresh your memory on the properties of logarithmic functions). The correct expression would be:

$$\begin{aligned} l_{\mathbf{X}}(\theta;\mathbf{x}) &= \log(f_{\mathbf{X}}(\theta;\mathbf{x})) \\ &= \log\left(\prod_{i=1}^{n} f_{X_i}(x_i;\theta)\right) \\ &= \sum_{i=1}^{n} \log f_{X_i}(x_i;\theta). \end{aligned}$$

Examination revision strategy

Many candidates are disappointed to find that their examination performance is poorer than they expected. This may be due to a number of reasons, but one particular failing is '**question spotting**', that is, confining your examination preparation to a few questions and/or topics which have come up in past papers for the course. This can have serious consequences.

We recognise that candidates might not cover all topics in the syllabus in the same depth, but you need to be aware that examiners are free to set questions on **any aspect** of the syllabus. This means that you need to study enough of the syllabus to enable you to answer the required number of examination questions.

The syllabus can be found in the Course information sheet available on the VLE. You should read the syllabus carefully and ensure that you cover sufficient material in preparation for the examination. Examiners will vary the topics and questions from year to year and may well set questions that have not appeared in past papers. Examination papers may legitimately include questions on any topic in the syllabus. So, although past papers can be helpful during your revision, you cannot assume that topics or specific questions that have come up in past examinations will occur again.

If you rely on a question-spotting strategy, it is likely you will find yourself in difficulties when you sit the examination. We strongly advise you not to adopt this strategy.

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Comments on specific questions

Candidates should answer all **FOUR** questions: **Question 1** of Section A (40 marks) and all **THREE** questions from Section B (60 marks in total). **Candidates are strongly advised to divide their time accordingly.**

Section A

Answer Question 1 from this section.

Question 1

(a) **Let $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ be a random sample from a distribution with the probability density function:**

$$f_X(x; \lambda, \theta) = \begin{cases} \lambda e^{-\lambda(x-\theta)} & \text{for } x \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

where $\lambda > 0$ and $\theta > 0$ are parameters.

i. **Find a 1-dimensional sufficient statistic for λ, when θ is known.**

(4 marks)

ii. **Find a 1-dimensional sufficient statistic for θ, when λ is known.**

(3 marks)

iii. **Find a 2-dimensional sufficient statistic for λ and θ.**

(3 marks)

Reading for this question

Section 2.8 of the subject guide covers sufficient statistics and the factorisation theorem.

Approaching the question

Due to independence, the joint density function of $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ is:

$$f_{\mathbf{X}}(\mathbf{x}; \lambda, \theta) = \begin{cases} \lambda^n \mathrm{e}^{-\lambda \sum_i x_i + n\lambda\theta} & \text{for } x_i \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

or else:

$$f_{\mathbf{X}}(\mathbf{x}; \lambda, \theta) = \lambda^n \mathrm{e}^{-\lambda \sum_i x_i} \mathrm{e}^{n\lambda\theta} I_{[\theta,\infty)}(x_{(1)}).$$

i. If θ is known, then we may set:

$$g\left(\sum_{i=1}^{n} x_i, \lambda\right) = \lambda^n \mathrm{e}^{-\lambda \sum_i x_i} \mathrm{e}^{n\lambda\theta} \quad \text{and} \quad h(\mathbf{x}) = I_{[\theta,\infty)}(x_{(1)}).$$

Since $f_{\mathbf{X}}(\mathbf{x}; \lambda, \theta) = g\left(\sum_{i=1}^{n} x_i, \lambda\right) h(\mathbf{x})$, the factorisation theorem implies that the statistic $\sum_{i=1}^{n} X_i$ is sufficient for λ.

ii. If λ is known, then we may set:

$$g\left(\min_i x_i, \theta\right) = \mathrm{e}^{n\lambda\theta}\, I_{[\theta,\infty)}(x_{(1)}) \quad \text{and} \quad h(\mathbf{x}) = \lambda^n \mathrm{e}^{-\lambda \sum_i x_i}.$$

Since $f_{\mathbf{X}}(\mathbf{x}; \lambda, \theta) = g(\min_i x_i, \theta)\, h(\mathbf{x})$, the factorisation theorem implies that the statistic $\min_i X_i = X_{(1)}$ is sufficient for θ.

iii. If neither θ nor λ are known, we may set:

$$g\left(\sum_{i=1}^{n} x_i, \min_i x_i, \lambda, \theta\right) = f_{\mathbf{X}}(\mathbf{x}; \lambda, \theta) \quad \text{and} \quad h(\mathbf{x}) = 1.$$

Since $f_{\mathbf{X}}(\mathbf{x}; \lambda, \theta) = g\left(\sum_{i=1}^{n} x_i, \min_i x_i, \lambda, \theta\right) h(\mathbf{x})$, the factorisation theorem implies that the statistic $\left(\sum_{i=1}^{n} X_i, X_{(1)}\right)$ is sufficient for (λ, θ).

(b) Let $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ be a sample from a distribution parameterised by θ. Let $T(\mathbf{X})$ be an unbiased estimator of a function of the parameter, denoted $g(\theta)$. Under regularity conditions permitting differentiation under an integral, for the *continuous* case show that for any $\theta \in \Theta$ it holds that:

$$\mathrm{Var}(T(\mathbf{X})) \geq \frac{\left(\frac{\mathrm{d}}{\mathrm{d}\theta} g(\theta)\right)^2}{\mathcal{I}_{\mathbf{X}}(\theta)}$$

where $\mathcal{I}_{\mathbf{X}}(\theta)$ is the total Fisher information about θ in $\mathbf{X}$.

***Hint*: You may use the variance/covariance form of the Cauchy–Schwarz inequality which states that for two random variables X and Y we have:**

$$(\mathrm{Cov}(X, Y))^2 \leq \mathrm{Var}(X)\,\mathrm{Var}(Y).$$

(10 marks)

Reading for this question

Section 3.8 of the subject guide covers the variance of unbiased estimators.

Approaching the question

By unbiasedness of $T(\mathbf{X})$, we have:

$$\mathrm{E}(T(\mathbf{X})) = g(\theta)$$

hence:

$$g(\theta) = \int_{\mathbb{R}^n} T(\mathbf{x})\, f_{\mathbf{X}}(\mathbf{x};\theta)\,\mathrm{d}\mathbf{x}.$$

Differentiating both sides with respect to θ gives us:

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\,g(\theta) = \int_{\mathbb{R}^n} T(\mathbf{x})\,\frac{\partial}{\partial\theta}\,f_{\mathbf{X}}(\mathbf{x};\theta)\,\mathrm{d}\mathbf{x} = \int_{\mathbb{R}^n} T(\mathbf{x})\,s_{\mathbf{X}}(\theta;\mathbf{x})\,f_{\mathbf{X}}(\mathbf{x};\theta)\,\mathrm{d}\mathbf{x} = \mathrm{E}(T(\mathbf{X})\,s_{\mathbf{X}}(\theta;\mathbf{X})).$$

Since $\mathrm{E}(s_{\mathbf{X}}(\theta;\mathbf{X})) = 0$, we have:

$$\begin{aligned}\mathrm{Cov}(T(\mathbf{X}), s_{\mathbf{X}}(\theta;\mathbf{X})) &= \mathrm{E}(T(\mathbf{X})\,s_{\mathbf{X}}(\theta;\mathbf{X})) - \mathrm{E}(T(\mathbf{X}))\,\mathrm{E}(s_{\mathbf{X}}(\theta;\mathbf{X}))\\ &= \mathrm{E}(T(\mathbf{X})\,s_{\mathbf{X}}(\theta;\mathbf{X}))\end{aligned}$$

hence we may write:

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\,g(\theta) = \mathrm{Cov}(T(\mathbf{X}), s_{\mathbf{X}}(\theta;\mathbf{X})).$$

Applying the Cauchy–Schwarz inequality then yields:

$$\left(\frac{\mathrm{d}}{\mathrm{d}\theta}\,g(\theta)\right)^2 \leq \mathrm{Var}(T(\mathbf{X}))\,\mathrm{Var}(s_{\mathbf{X}}(\theta;\mathbf{X})).$$

Rearranging, we have:

$$\mathrm{Var}(T(\mathbf{X})) \geq \frac{\left(\frac{\mathrm{d}}{\mathrm{d}\theta}\,g(\theta)\right)^2}{\mathrm{Var}(s_{\mathbf{X}}(\theta;\mathbf{X}))} = \frac{\left(\frac{\mathrm{d}}{\mathrm{d}\theta}g(\theta)\right)^2}{\mathcal{I}_{\mathbf{X}}(\theta)}$$

since $\mathcal{I}_{\mathbf{X}}(\theta) = \mathrm{Var}(s_{\mathbf{X}}(\theta;\mathbf{X}))$.

(c) Let $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ be a random sample from a Geo(π) distribution, with the probability mass function:

$$p_X(x;\pi) = \begin{cases}(1-\pi)^{x-1}\pi & \text{for } x = 1, 2, \ldots\\ 0 & \text{otherwise}\end{cases}$$

for $0 < \pi < 1$. Derive the Fisher information for π.

Hint: **You may use the fact that if $X \sim$ Geo(π), then $\mathrm{E}(X) = 1/\pi$.**

(10 marks)

Reading for this question

Section 2.10.2 of the subject guide covers the score function and Fisher information.

Approaching the question

Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\pi;\mathbf{x}) = \prod_{i=1}^{n}(1-\pi)^{x_i-1}\pi = (1-\pi)^{-n+\sum_i x_i}\pi^n.$$

Hence the log-likelihood function is:

$$l_{\mathbf{X}}(\pi;\mathbf{x}) = \log L_{\mathbf{X}}(\pi;\mathbf{x}) = \left(-n + \sum_{i=1}^{n} x_i\right)\log(1-\pi) + n\log(\pi).$$

Differentiating:

$$\frac{\partial}{\partial\pi}\,l_{\mathbf{X}}(\pi;\mathbf{x}) = -\frac{-n+\sum_i x_i}{1-\pi} + \frac{n}{\pi}$$

and:

$$\frac{\partial^2}{\partial\pi^2}\,l_{\mathbf{X}}(\pi;\mathbf{x}) = -\frac{-n+\sum_i x_i}{(1-\pi)^2} - \frac{n}{\pi^2}.$$

Fisher information is:

$$I_{\mathbf{X}}(\pi) = -\mathrm{E}\left[\frac{\partial^2}{\partial\pi^2}\,l_{\mathbf{X}}(\pi;\mathbf{x})\right] = \frac{-n+\sum_i \mathrm{E}(X_i)}{(1-\pi)^2} + \frac{n}{\pi^2} = \frac{n(1-\pi)/\pi}{(1-\pi)^2} + \frac{n}{\pi^2} = \frac{n}{\pi^2(1-\pi)}.$$

(d) **Let $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ be a random sample from the $N(0, \sigma^2)$ distribution. Find a pivotal function for the construction of a confidence interval for σ^2. Use the pivotal function to construct an exact $100(1-\alpha)\%$ confidence interval for σ^2.**

(10 marks)

Reading for this question

Section 4.6 of the subject guide covers finding interval estimators from pivotal functions.

Approaching the question

Since each $X_i \sim N(0, \sigma^2)$, then each $X_i/\sigma \sim N(0,1)$. Due to independence, we have that:

$$\frac{\sum_{i=1}^{n} X_i^2}{\sigma^2} \sim \chi^2_n$$

which is a pivotal function for σ^2.

Let $\chi^2_{\alpha,\,n}$ denote the top 100αth percentile of a χ^2_n distribution. As $\sum_{i=1}^{n} X_i^2/\sigma^2 \sim \chi^2_n$, we can write:

$$1-\alpha = P\left(\chi^2_{1-\alpha/2,\,n} \le \frac{\sum_{i=1}^{n} X_i^2}{\sigma^2} \le \chi^2_{\alpha/2,\,n}\right) = P\left(\frac{\sum_{i=1}^{n} X_i^2}{\chi^2_{\alpha/2,\,n}} \le \sigma^2 \le \frac{\sum_{i=1}^{n} X_i^2}{\chi^2_{1-\alpha/2,\,n}}\right)$$

suggesting that an exact $100(1-\alpha)\%$ confidence level interval for σ^2 is:

$$\left[\frac{\sum_{i=1}^{n} X_i^2}{\chi^2_{\alpha/2,\,n}}, \frac{\sum_{i=1}^{n} X_i^2}{\chi^2_{1-\alpha/2,\,n}}\right].$$

Section B

Answer all three questions from this section.

Question 2

Let $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$. It is required to estimate the parameter μ, where $\mu > 0$. It is thought that the sample mean, $\widehat{\mu} = \bar{X}$, is the best estimator of μ, however someone proposes an alternative estimator, $\widetilde{\mu}$, which requires an unbiased coin to be tossed such that:

$$\widetilde{\mu} = \begin{cases} \bar{X} & \text{if the unbiased coin comes up heads} \\ 0 & \text{if the unbiased coin comes up tails.} \end{cases}$$

Note that in this question you may state without proof any known properties of $\bar{X}$.

(a) **Show that $\widetilde{\mu}$ can be represented as:**

$$\widetilde{\mu} = \varepsilon \bar{X}$$

where ε is a random variable independent of $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ such that:

$$P(\varepsilon = 0) = P(\varepsilon = 1) = \frac{1}{2}.$$

(4 marks)

(b) Determine whether $\widetilde{\mu}$ is a biased estimator of μ.

(4 marks)

(c) Derive the variance of $\widetilde{\mu}$.

(6 marks)

(d) By comparing their respective mean squared errors, decide which of $\widehat{\mu}$ and $\widetilde{\mu}$ is the better estimator of μ.

(6 marks)

Reading for this question

Section 3.4 of the subject guide covers the statistical properties of point estimators, such as bias, variance and mean squared error.

Approaching the question

(a) Let ε be a random variable such that:

$$\varepsilon = \begin{cases} 1 & \text{if the unbiased coin comes up heads} \\ 0 & \text{if the unbiased coin comes up tails} \end{cases}$$

which is obviously independent of the sample $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$. Hence by the construction of $\widetilde{\mu}$ we have:

$$\widetilde{\mu} = (1-\varepsilon) \times 0 + \varepsilon \bar{X} = \varepsilon \bar{X}.$$

(b) Due to independence of ε and $\bar{X}$, we have:

$$\text{Bias}(\widetilde{\mu}) = \text{E}(\widetilde{\mu}) - \mu = \text{E}(\varepsilon \bar{X}) - \mu = \text{E}(\varepsilon)\,\text{E}(\bar{X}) - \mu = \frac{1}{2}\,\mu - \mu = -\frac{\mu}{2} < 0$$

hence $\widetilde{\mu}$ is a negatively biased estimator of μ (since $\mu > 0$).

(c) Due to independence of ε and $\bar{X}$, we have:

$$\begin{aligned} \text{Var}(\widetilde{\mu}) = \text{E}(\widetilde{\mu}^2) - (\text{E}(\widetilde{\mu}))^2 &= \text{E}(\varepsilon^2 \bar{X}^2) - \left(\frac{\mu}{2}\right)^2 \\ &= \text{E}(\varepsilon^2)\,\text{E}(\bar{X}^2) - \frac{\mu^2}{4} \\ &= \text{E}(\varepsilon)(\text{Var}(\bar{X}) + (\text{E}(\bar{X}))^2) - \frac{\mu^2}{4} \\ &= \frac{1}{2}\left[\frac{\sigma^2}{n} + \mu^2\right] - \frac{\mu^2}{4} \\ &= \frac{\sigma^2}{2n} + \frac{\mu^2}{4}. \end{aligned}$$

(d) Noting that $\text{Bias}(\widehat{\mu}) = 0$ and $\text{Bias}(\widetilde{\mu}) = -\mu/2$, we have:

$$\text{MSE}(\widehat{\mu}) = \text{Var}(\widehat{\mu}) + (\text{Bias}(\widehat{\mu}))^2 = \frac{\sigma^2}{n}$$

and:

$$\text{MSE}(\widetilde{\mu}) = \text{Var}(\widetilde{\mu}) + (\text{Bias}(\widetilde{\mu}))^2 = \frac{\sigma^2}{2n} + \frac{\mu^2}{4} + \left(-\frac{\mu}{2}\right)^2 = \frac{\sigma^2}{2n} + \frac{\mu^2}{2}.$$

Comparing mean squared errors, then $\text{MSE}(\widetilde{\mu}) < \text{MSE}(\widehat{\mu})$ if:

$$\frac{\sigma^2}{2n} + \frac{\mu^2}{2} < \frac{\sigma^2}{n} \quad \Rightarrow \quad \frac{\mu^2}{2} < \frac{\sigma^2}{2n} \quad \Rightarrow \quad \mu < \frac{\sigma}{\sqrt{n}}.$$

In other words if we know that $\mu > 0$ is sufficiently 'small', it makes sense to use the estimator $\widetilde{\mu}$ rather than $\widehat{\mu}$ since its mean squared error is smaller.

Question 3

Suppose $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ is a random sample from the distribution with the probability density function:

$$f_X(x;\theta) = \begin{cases} 4x^3/\theta^4 & \text{for } 0 \leq x \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

where $\theta > 0$ is a parameter. Let $X_{(i)}$ be the ith smallest value in the sample, for $i = 1, 2, \ldots, n$.

(a) **Find the probability density functions of the sample maximum *and* the sample minimum.**

(6 marks)

(b) **Find a simple sufficient statistic for θ.**

(3 marks)

(c) **Determine the maximum likelihood estimator of θ and show that it is a biased estimator of θ.**

(5 marks)

(d) **Use your estimator found in part (c) to find an unbiased estimator of θ and compute its mean squared error. Briefly explain whether this estimator is mean-square consistent.**

(6 marks)

Reading for this question

Section 2.8 of the subject guide covers sufficient statistics, and Section 3.9 covers maximum likelihood estimation.

Approaching the question

(a) For $x \in [0, \theta]$, we have:

$$F_X(x;\theta) = \int_0^x \frac{4t^3}{\theta^4}\,\mathrm{d}t = \left[\left(\frac{t}{\theta}\right)^4\right]_0^x = \left(\frac{x}{\theta}\right)^4.$$

Also:

$$F_{X_{(n)}}(x;\theta) = P(X_{(n)} \leq x) = P(X_i \leq x \text{ for all } i) = (P(X_1 \leq x))^n = (F_X(x;\theta))^n.$$

Hence the probability density function of the sample maximum is:

$$f_{X_{(n)}}(x;\theta) = n(F_X(x;\theta))^{n-1} f_X(x;\theta) = \frac{4nx^{4n-1}}{\theta^{4n}}\, I_{[0,\theta]}(x).$$

For $x \in [0, \theta]$, we have:

$$\begin{aligned} F_{X_{(1)}}(x;\theta) = P(X_{(1)} \leq x) = 1 - P(X_{(1)} > x) &= 1 - P(X_i > x \text{ for all } i) \\ &= 1 - (P(X_1 > x))^n \\ &= 1 - (1 - F_X(x;\theta))^n. \end{aligned}$$

Hence the probability density function of the sample minimum is:

$$f_{X_{(1)}}(x;\theta) = n(1 - F_X(x;\theta))^{n-1} f_X(x;\theta) = \frac{4nx^3(\theta^4 - x^4)^{n-1}}{\theta^{4n}}\, I_{[0,\theta]}(x).$$

(b) Due to independence, the joint probability density function is:

$$f_{\mathbf{X}}(\mathbf{x};\theta) = \prod_{i=1}^{n} f_X(x_i;\theta) = \prod_{i=1}^{n} \frac{4x_i^3}{\theta^4}\, I_{[0,\theta]}(x_i)$$

$$= \frac{4^n}{\theta^{4n}} \prod_{i=1}^{n} x_i^3 \prod_{i=1}^{n} I_{[0,\theta]}(x_i)$$

$$= \frac{4^n}{\theta^{4n}} \prod_{i=1}^{n} x_i^3 I_{[0,\infty)}(x_{(1)})\, I_{[0,\theta]}(x_{(n)}).$$

Hence, by the factorisation theorem, $X_{(n)}$ is sufficient for θ.

(c) The likelihood function $L_{\mathbf{X}}(\theta;\mathbf{x}) \equiv f_{\mathbf{X}}(\mathbf{x};\theta)$ is a decreasing function of θ when $\theta \in [X_{(n)},\infty)$, and $L_{\mathbf{X}}(\theta;\mathbf{x}) = 0$ when $\theta < X_{(n)}$. Therefore, the maximum likelihood estimator is $\widehat{\theta} = X_{(n)}$. Since:

$$\mathrm{E}(\widehat{\theta}) = \mathrm{E}(X_{(n)}) = \int_{-\infty}^{\infty} x\, f_{X_{(n)}}(x;\theta)\,\mathrm{d}x = \int_0^{\theta} \frac{4nx^{4n}}{\theta^{4n}}\,\mathrm{d}x = \left[\frac{4nx^{4n+1}}{(4n+1)\theta^{4n}}\right]_0^{\theta} = \frac{4n}{4n+1}\,\theta \neq \theta$$

then $\widehat{\theta}$ is a negatively biased estimator of θ.

(d) Let $\widetilde{\theta} = (4n+1)X_{(n)}/4n$. Hence $\widetilde{\theta}$ is an unbiased estimator of θ because:

$$\mathrm{E}(\widetilde{\theta}) = \mathrm{E}\left(\frac{4n+1}{4n}\,X_{(n)}\right) = \frac{4n+1}{4n}\,\frac{4n}{4n+1}\,\theta = \theta.$$

Hence:

$$\mathrm{MSE}(\widetilde{\theta}) = \mathrm{Var}(\widetilde{\theta}) + (\mathrm{Bias}(\widetilde{\theta}))^2 = \mathrm{Var}\left(\frac{4n+1}{4n}\,X_{(n)}\right) + 0^2 = \left(\frac{4n+1}{4n}\right)^2 \mathrm{Var}(X_{(n)}).$$

We have:

$$\mathrm{E}(X_{(n)}^2) = \int_{-\infty}^{\infty} x^2 f_{X_{(n)}}(x;\theta)\,\mathrm{d}x = \int_0^{\theta} \frac{4nx^{4n+1}}{\theta^{4n}}\,\mathrm{d}x = \left[\frac{4nx^{4n+2}}{(4n+2)\theta^{4n}}\right]_0^{\theta} = \frac{4n}{4n+2}\,\theta^2.$$

Hence:

$$\mathrm{Var}(X_{(n)}) = \mathrm{E}(X_{(n)}^2) - (\mathrm{E}(X_{(n)}))^2 = \frac{4n}{4n+2}\,\theta^2 - \left(\frac{4n}{4n+1}\right)^2 \theta^2 = \frac{4n}{(4n+2)(4n+1)^2}\,\theta^2.$$

Therefore:

$$\mathrm{MSE}(\widetilde{\theta}) = \left(\frac{4n+1}{4n}\right)^2 \frac{4n}{(4n+2)(4n+1)^2}\,\theta^2 = \frac{\theta^2}{4n(4n+2)}$$

which $\to 0$ as $n \to \infty$, therefore $\widetilde{\theta}$ is mean-square consistent.

Question 4

(a) **State the Neyman–Pearson lemma.**

(5 marks)

Reading for this question

Section 5.5.1 of the subject guide presents the Neyman–Pearson lemma.

Approaching the question

Consider a test procedure T of size α for $\mathrm{H}_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ vs. $\mathrm{H}_1 : \boldsymbol{\theta} = \boldsymbol{\theta}_1$. If the rejection region for T is:

$$R_T = \{\mathbf{x} \in \mathbb{R}^n : L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x}) - k_\alpha L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x}) > 0\}$$

equivalently:

$$R_T = \left\{\mathbf{x} \in \mathbb{R}^n : \frac{L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x})}{L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x})} > k_\alpha\right\}$$

then T is the most powerful test of size α.

Here $L_{\mathbf{X}}$ is the likelihood function and k_α is a constant dependent on α. Note that $\alpha = P_{\boldsymbol{\theta}_0}(\mathbf{X} \in R_T)$.

(b) Let $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ be a random sample from a distribution with the following probability density function:

$$f_X(x; \beta) = \begin{cases} \beta x^{\beta - 1} & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

where $\beta > 0$ is a parameter.

i. Consider the test of:

$$\mathrm{H}_0 : \beta = \beta_0 \quad \text{vs.} \quad \mathrm{H}_1 : \beta \neq \beta_0.$$

Derive the likelihood ratio test statistic for this hypothesis test.

(6 marks)

ii. Construct an asymptotic test of size α, for $\alpha = 0.10$, 0.05 and 0.01.

***Hint:* The table below may be useful.**

Degrees of freedom, ν	$\chi^2_{0.10,\,\nu}$	$\chi^2_{0.05,\,\nu}$	$\chi^2_{0.025,\,\nu}$	$\chi^2_{0.01,\,\nu}$
1	**2.706**	**3.841**	**5.024**	**6.635**
2	**4.605**	**5.991**	**7.378**	**9.210**
3	**6.251**	**7.815**	**9.348**	**11.345**
4	**7.779**	**9.488**	**11.143**	**13.277**
5	**9.236**	**11.070**	**12.832**	**15.086**

(4 marks)

iii. Describe how you would obtain an asymptotic p-value for the test in part ii., and provide a graphical representation of it.

(5 marks)

Reading for this question

Section 5.6 of the subject guide presents likelihood ratio tests, while Section 5.4.3 covers p-values.

Approaching the question

i. Due to independence, the likelihood function for this problem is:

$$L_{\mathbf{X}}(\beta; \mathbf{x}) = \prod_{i=1}^{n} \beta x_i^{\beta - 1} = \beta^n \left(\prod_{i=1}^{n} x_i\right)^{\beta - 1}.$$

The likelihood ratio test statistic, $LR(\mathbf{X})$, for this test can be written as:

$$LR(\mathbf{X}) = \frac{L_{\mathbf{X}}(\widehat{\beta}; \mathbf{x})}{L_{\mathbf{X}}(\beta_0; \mathbf{x})}$$

where $\widehat{\beta}$ is the maximum likelihood estimator (MLE) of β. To find $\widehat{\beta}$ consider the log-likelihood function which is:

$$l_{\mathbf{X}}(\beta;\mathbf{x}) = \log L_{\mathbf{X}}(\beta;\mathbf{x}) = n\log\beta + (\beta-1)\sum_{i=1}^{n}\log x_i$$

and the score function is:

$$s_{\mathbf{X}}(\beta;\mathbf{X}) = \frac{\partial}{\partial\beta}\, l_{\mathbf{X}}(\beta;\mathbf{x}) = \frac{n}{\beta} + \sum_{i=1}^{n}\log x_i.$$

Setting $s_{\mathbf{X}}(\beta;\mathbf{X}) = 0$ leads to $\widehat{\beta} = -n/\sum_i \log X_i$ as a candidate MLE. Since:

$$\frac{\partial^2}{\partial\beta^2}\, l_{\mathbf{X}}(\beta;\mathbf{x}) = -\frac{n}{\beta^2} < 0$$

we conclude that this is indeed the MLE. Hence we can write $LR(\mathbf{X})$ as:

$$LR(\mathbf{X}) = \frac{\widehat{\beta}^n \left(\prod_{i=1}^{n} x_i\right)^{\widehat{\beta}-1}}{\beta_0^n \left(\prod_{i=1}^{n} x_i\right)^{\beta_0-1}} = \frac{\left(-\frac{n}{\sum_i \log x_i}\right)^n \left(\prod_{i=1}^{n} x_i\right)^{-\frac{n}{\sum_i \log x_i}-1}}{\beta_0^n \left(\prod_{i=1}^{n} x_i\right)^{\beta_0-1}}.$$

ii. For large n, the distribution of $2\log LR(\mathbf{X})$ is χ^2_1 under H_0, with large values of $2\log LR(\mathbf{X})$ being against H_0. Hence a size α test will reject H_0 if $2\log LR(\mathbf{X}) > \chi^2_{\alpha,\,1}$. For a 10% significance level, the critical value would be $\chi^2_{0.10,\,1} = 2.706$. For a 5% significance level, the critical value would be $\chi^2_{0.05,\,1} = 3.841$. For a 1% significance level, the critical value would be $\chi^2_{0.01,\,1} = 6.635$.

iii. For the asymptotic test of part ii. we can use the observed value of $LR(\mathbf{x}) = x$, say, and the fact that $2\log LR(\mathbf{X}) \sim \chi^2_{\alpha,\,1}$ under H_0, to calculate the following probability:

$$p\text{-value } = P_{\mathrm{H}_0}(2\log LR(\mathbf{X}) > x) = P(\chi^2_{\alpha,\,1} > x).$$

Candidates are also expected to provide a rough sketch of the area corresponding to the p-value.